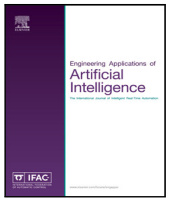
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Survey paper

A comprehensive survey and taxonomy of sign language research



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ABSTRACT

Sign language relies on visual gestures of human body parts to convey meaning and plays a vital role in modern society to communicate and interact with people having hearing difficulty as well as for human–machine interaction applications. This field has attracted a growing attention in recent years and several research outcomes have been witnessed covering various issues including sign acquisition, segmentation, recognition, translation and linguistic structures. In this paper, a comprehensive up-to-date survey of the state-of-the-art literature of automated sign language processing is presented. The survey provides a taxonomy and review of the body of knowledge and research efforts with focus on acquisition devices, available databases, and recognition techniques for fingerspelling signs, isolated sign words, and continuous sentence recognition systems. It covers recent advances including deep machine learning and multimodal approaches and discusses various related challenges. This survey is directed to junior researchers and industry developers working on sign language gesture recognition and related systems to gain insights and identify distinctive aspects and current status of existing landscape as well as future perspectives leading to further advancements.

1. Introduction

Communication plays an essential role with tremendous effects on human life, e.g. gain and exchange knowledge, interact and develop social relationships, and express feelings and needs. While spoken language is the normal way for people to communicate, sign language (SL) represents a main alternative for people with auditory or vocal impairment or difficulty. Recently, the World Health Organization (WHO) declared that more than 400 million people worldwide have disabling hearing loss which forms about 5% of the world's population.[1] Hearing impairment may range from mild (20 dB–40 dB below normal hearing threshold) to profound (more than 81 dB below normal hearing threshold) and can be caused as a result of a birth defect or an illness, or develop at a later stage of age. With the recent advances in information technology and robotics, sign gestures is not only important to overcome communication barriers between humans but also for human–machine interaction (Ryumin et al., 2019; Homburg et al., 2019) and translation of sign language into another form of natural language and vice versa (Dutta et al., 2015; Rajaganapathy et al., 2015).

Though there are some references to a sign language that dates back to the 5th century BC, it was not until the past centuries where some attempts have been made to develop a system for deaf and dump communication within their community and to other people. For instance, an early attempt in western countries in 16th century attributed to a Spanish monk, Pedro Ponce de León, as the first teacher for the deaf who described a manual alphabet for fingerspelling, formed the base for Juan Pablo Bonet's writings in 1620 for teaching deaf people. The first free school for deaf was founded in Paris in 1755 and the first American School for deaf was founded in Connecticut in 1817. Almost in parallel, the British sign language has evolved and widely varied from the American sign language. However, a resolution was passed in Milan in 1880 at the International Congress on Education of the Deaf to officially prohibit using sign language in favor of speech. Not until the past two decades, some countries such as UK have recognized the sign language as a language in its own right and become part of the formal educational system rather than charity boards. Similar to natural languages, sign language is not universal, i.e. it is geographically specific and more than 135 variants exist around the world (Luqman et al., 2017; Makhashen et al., 2019). Examples of popular sign languages include Spanish (Lengua de Signos Española or LSE), American (ASL), British (BSL), Irish (ISL), French (langue des signes française or LSF), Australian (Auslan), Chinese (CSL), Dutch (Nederlandse Gebarentaal or NGT), German (Deutsche Gebärdensprache or DGS)), Brazilian (Libras),

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[1] https://www.who.int/news-room/fact-sheets/detail/deafness-and-hearing-loss.

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Fig. 1. Different facial expressions used with (a) "Beautiful" and (b) "Ugly" signs.

Russian (RSL) and Arabic (ArSL). Moreover, some countries may share the same sign language alphabet, e.g. British, Australian and New Zealand (BANZSL), while others can have different sign languages albeit sharing the same spoken language, e.g. BSL and ASL.

Sign language is considered as a spontaneous non-verbal way for people to communicate through manual and/or non-manual body gestures for visual signing to convey meaning. Manual gestures are the primary components of a sign language and usually involve hand movements to express the signer's emotions and thoughts (Cheok et al., 2019). Manual gestures are naturally accompanied with simultaneous non-manual gestures such as head movements, facial expressions and body postures (Nair and Bindu, 2013). Facial expressions employ different face parts such as eye, eyebrow, forehead, and mouth to express or stress meanings that cannot be revealed by manual gestures. Facial expressions are a central focal point to mark syntactical and grammatical information in sign languages such as conditions and questions. For instance, they can be used to express negation or serve as adverbs and adjectives of manual signs distinguishing between signs with similar hand gestures (Goldstein and Feldman, 1996). An example is shown in Fig. 1 where facial expressions are used to reflect the emotion of two ArSL signs: "Beautiful" and "Ugly". Each sign in the first group shares the face posture with a corresponding similar sign in the second group. Moreover, head motion is an important component of the non-manual articulators of a sign language that can be integrated with manual gestures or used as an independent sign. Fig. 2 shows an illustrative example of a sign that uses hand and head postures together with eye closing to express the "Sleep" sign in ArSL. Lip motion is another popular form of non-manual gestures for voiced and unvoiced speech recognition but recently is gaining increasing attention for sign language recognition (Koller et al., 2015b). Sign gestures can be classified differently into static or dynamic gestures. The former type keeps the position of the body part fixed during signing and typically depends on the orientation and shape of hands and fingers for signing digits and alphabets (Pisharady and Saerbeck, 2015). In contrast, dynamic gestures depend on hands and body motion during signing and this type of gestures is used mainly in signing words and sentences (Luqman and Mahmoud, 2019).

Automated Sign Language Recognition (SLR) continues to gain interest within the research community and in recent years there has been an influx of related publications. While a number of surveys and reviews were published, they mostly focus on specific issues and the continuous development of the field necessitates a more up-to-date literature survey. This paper aims at providing a comprehensive survey and taxonomy for the state-of-the-art sign language recognition. It explores various aspects and techniques related to sign language. Furthermore, it covers latest advances including deep learning and reviews most common resources available for sign language recognition.



Fig. 2. "Sleep" sign in ArSL combining hand, head, and eye gestures.

It describes publicly available datasets used in the related research. Moreover, it discusses existing challenges and highlights potential research directions for sign language recognition. This survey is directed to junior researchers and industry developers in related fields to gain key insights of sign language recognition and related human–machine interaction systems. The remainder of the paper is organized as follows: Section 2 provides an overview and reviews related works. Section 3 describes different sign acquisition devices while Section 4 lists available datasets for sign language recognition. Section 5 extensively discusses different techniques adopted for sign language recognition. Finally, conclusions are directions for advancing the field are outlined in Section 6.

2. Overview and existing reviews

Sign language recognition has attracted the attention of researchers from multidisciplinary fields including computer vision, pattern recognition, machine learning, sensors and robotics. Researchers have explored different issues in order to develop real-time reliable systems to communicate effectively with hearing or vocal impaired people. The recent advancement in computer vision and deep learning has fueled research efforts in sign language recognition. In Fig. 3, we show a mindmap taxonomy of various research directions related to specific languages or cross-language approaches. Sign language recognition can be classified based on body gestures (manual or non-manual) (Koller et al., 2015b; Aran et al., 2007a; Caridakis et al., 2014; Xu et al., 2000; Dutta et al., 2015), acquisition methods (vision, sensor, depth, or hybrid) (Zaki and Shaheen, 2011; Aloysius and Geetha, 2020; Mittal et al., 2019; Ahmed et al., 2018; Ghotkar and Kharate, 2015; Yang, 2015), sign category (fingerspelling, isolated, or continuous) (Starner et al., 1998; Wang et al., 2016b), detection methods (static/non-tracking, dynamic/tracking, or segmentation) (Liao et al., 2019), recognition

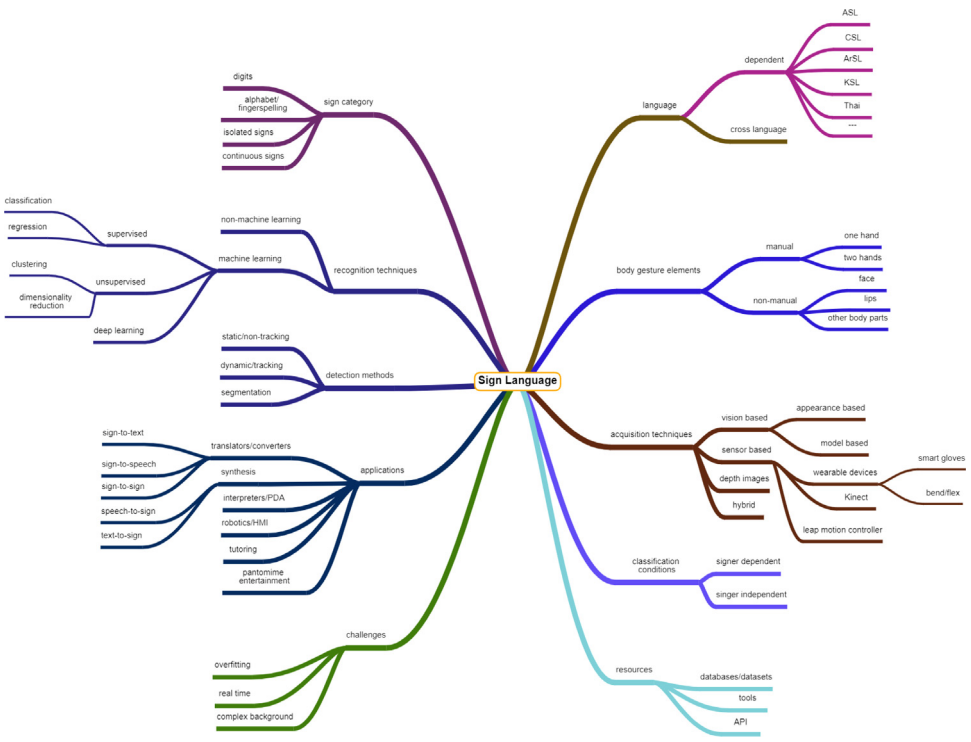


Fig. 3. Taxonomy of research directions on sing language related issues.

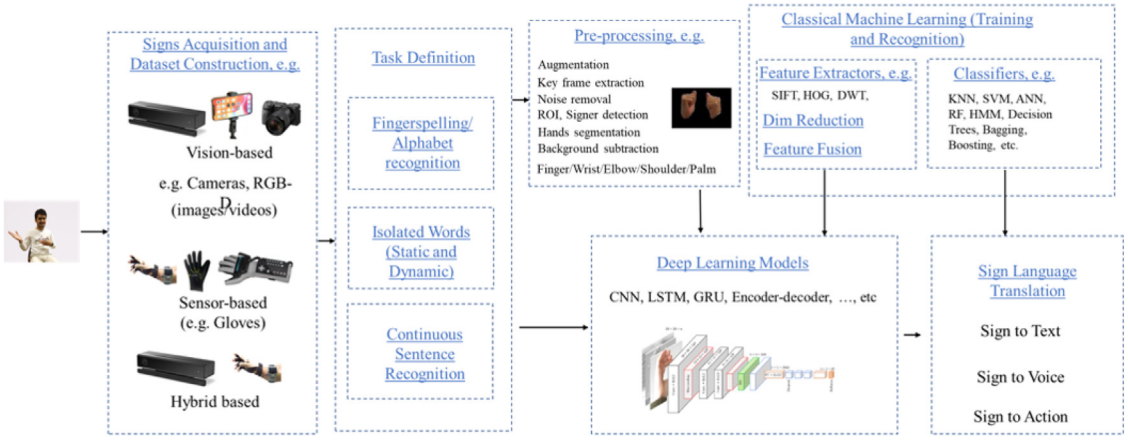


Fig. 4. Workflow of sign language recognition and translation.

techniques (machine learning or non-machine learning) (Ong and Ranganath, 2005), and classification conditions (signer-dependent or signer independent) (Zieren and Kraiss, 2005; Aly et al., 2019). Some researchers were also interested on handling special issues related to data overfitting, real time, and complex background. Others developed tools and datasets to advance the research in the field. Currently there are various applications of sign language systems such as sign synthesis or generation (text-to-sign and speech-to-sign) (Stoll et al., 2020; Zelinka and Kanis, 2020; Xiao et al., 2020), sign translation or conversion (sign-to-sign, sign-to-text, or sign-to-speech) (Camgoz et al., 2018; Luqman and Mahmoud, 2019; Ko et al., 2019; Guo et al., 2019; Khan et al., 2016; Song et al., 2019; Abiyev et al., 2020; Zhou et al., 2020; Jose and Julian, 2019), sign language tutoring (Aran et al., 2009; Potamianos et al., 2020), and human–machine interaction (Zieren and Kraiss, 2004; Homburg et al., 2019). Fig. 4 shows a workflow for sign language recognition and translation based on vision, sensor or hybrid approaches.

Although there have been some efforts to concisely survey published research on sign gestures and recognition, most of these surveys are not either up-to-date or cover limited issues in the field. The first comprehensive attempt (Ong and Ranganath, 2005) surveyed work before 2005 related to automatic analysis of manual signing. Later in 2011, a conference paper (Kausar and Javed, 2011) came out to summarize 17 papers published between 2005 and 2011. A brief review of 33 papers published between 1999 and 2013 is presented in Mohandes et al. (2014b,a) with focus on image-based systems and methods for Arabic sign language. Some researchers addressed special issues for sign languages, e.g. sensory gloves (Ahmed et al., 2018) and mobile technology (Neiva and Zanchettin, 2018). Ahmed et al. (2018) provided a review of glove-based systems by analyzing 71 articles collected from Web of Science, IEEE and ScienceDirect and related to several aspects of sign-language recognition, with 48 papers related to one hand gesture while only 7 papers were on both hands. Due to the increasing progress of mobile technology and to overcome the barrier between deaf and hearing persons, a review of work related to mobile technology for

sign languages is provided in Neiva and Zanchettin (2018). It covers 43 papers from 2009 to 2017 for gesture and facial recognition in the context of sign languages. The reviewed papers are classified based on techniques for feature extraction, static gesture recognition, dynamic gesture recognition, facial expression, gaze gesture recognition and special hardware for gesture recognition.

A relatively more comprehensive survey was published in 2016 but concentrated on Indian sign language (Agrawal et al., 2016). It reviewed techniques for hand segmentation and tracking, feature extraction, and classification for both manual and non-manual, isolated and continuous, glove and device based, signer dependence and independence. Similarly, two other surveys were recently published on Chinese sign language (Jiang et al., 2020; Kamal et al., 2019). Two more informative surveys are presented in Cheok et al. (2019), Galván-Ruiz et al. (2020). The former paper was published online in 2017 and reviewed papers up to 2016. Though the paper is on sign language, most techniques are related to hand gestures, which the authors categorized as conversational gestures, controlling gestures, manipulative gestures and communicative gestures. Sign language is a type of the latter form of hand gestures but highly structured with considerably large intra-class variations. Galván-Ruiz et al. (2020) presented a timeline analysis of work related to gesture acquisition technologies with applications to sign languages including gloves, electromyography electrodes, ultrasound, WiFi, RFID, RGB and RGBD cameras, leap motion, and virtual reality.

A systematic literature review of prominent recognition techniques, applications and challenges for the general subject of hand gestures is presented in Yasen and Jusoh (2019). It covers 95 selected papers from IEEE Explore between 2016 and 2018. This work reported that the dominant techniques in the analyzed papers are wearable hand gesture devices for data acquisition and artificial neural networks for classification. Another systematic review is presented in Wadhawan and Kumar (2019) where 117 papers published between 2007 and 2017 were selected and categorized based on 25 sign languages. The analysis of these selected papers demonstrated that the main research areas on sign language are related to static isolated single hand signs. Recently, some modest reviews are presented, e.g. 30 papers are analyzed in Er-Rady et al. (2017) covering various issues related to sign language recognition, and 21 papers on vision-based sign language recognition techniques are reviewed in Sharma and Singh (2020). Another brief review for non-manual gestures is presented in Antonakos et al. (2015) but concentrates on mouth motion modeling as an important component for sign language recognition.

Unlike previous reviews, this paper provides a more comprehensive survey and taxonomy of various research directions pertaining to different sign languages, acquisition devices, recognition methods, classification conditions, resources and challenges. The following electronic libraries were selected as data sources for our study: ACM Digital Library, Scopus, Springer, IEEE-Xplore, and Google Scholar. The search keywords are: Sign Language Recognition, Sign Language Translation, Sign Language Database, Sign Language Dataset, Continuous Sign Language, and Gesture Recognition + Sign Language. We considered relevant publications until 2021 to cover various research directions.

3. Sign acquisition devices

Acquisition techniques of sign language can be classified into sensor-based, vision-based, and a combination of the two (Ahmed et al., 2018). Sensor-based acquisition techniques depend on a set of sensors to record the signers' gesture information (Ahmed et al., 2018; Zhang et al., 2019b) while vision-based techniques use video cameras to capture signers' gesture (Pisharady and Saerbeck, 2015).



Fig. 5. Samples of electronic gloves used for sign language acquisition (a) CyperGoves (Mohandes, 2013), (b) DG5-Vhand (Sadek et al., 2017), and (c) PowerGlove (Mohandes et al., 2004).

3.1. Sensor-based acquisition techniques

Sensor-based techniques employ a set of sensors to capture the hand configurations and represent them as features. This type of acquisition techniques requires the signer to wear cumbersome gloves during signing. These gloves are used to track and record hands' and fingers' motion. They provide information on the position, rotation, movement, orientation of the hand, and more importantly, finger bending. Samples of these glove sensors are shown in Fig. 5.

The sensors used for sign language acquisition can be classified into sensors to detect finger bending and sensors to detect hand motion and orientation (Ahmed et al., 2018). Fingers detection sensors are used to detect and track hand fingers movement. Flex, optical, and tactile sensors are examples of this type of sensors (Ahmed et al., 2018). Hand motion sensors detect the movement and the orientation of the hand in addition to having the capability of detecting fingers movement. For more information about sensors used for sign language, we refer to this survey paper (Ahmed et al., 2018).

Several techniques in the literature utilized sensors information for sign language recognition. Most of these techniques used either gloves alone (Liang and Ouhyoung, 1998; Kong and Ranganath, 2014; Ritchings et al., 2012; Mohandes and Deriche, 2013) or used gloves with hand tracker sensors (Gao et al., 2004b; Wang et al., 2001; Fang et al., 2001; Yao et al., 2006). Other techniques employed other sensors that provide data (accelerometer and gyroscope) about the performed signs such as Smartwatches sensors (Ekiz et al., 2017) and wearable IMU system (Suri and Gupta, 2019).

Ritchings et al. developed a computer-based system using Data Glove for teaching sign language (Ritchings et al., 2012). Bend sensors and push button switches were utilized to observe 17 measures. There are two bend sensors joined to every finger and thumb junctions to monitor the finger movements. Two bend sensors on the inner and on the outer side of the hand wrist to catch its up–down movement. In addition, a bend sensor on the right side of the wrist to sense its yaw. The bend sensors reports different measures when the bent is in one direction, which is suitable for the fingers. Two sensors are mounted on opposite sides of the wrist to monitor pitch up/down movements. Lastly, to identify finger abduction and adduction, four push-down switches are attached to the side of every finger beginning with the thumb. The focus of the system was on evaluating the capability of learners in reproducing signs done by a professional signer. The used database covered 65 signs performed by four professional signers. All the signs were selected to reflect all the activities that could be done by a learner within the bounds of the 17 sensor configurations. The learners were able to reproduce the signs with an accuracy of 93%. They also reported that their system faced difficulties in recognizing some signs made by a trainee with a small hand.

Mohandes and Deriche (2013) used the Dempster–Shafer (DS) Theory of Evidence to combine decisions from a Cyber Glove with 22 sensors and a hand tracking system. The system is used to recognize 20 two-handed signs. Each glove provides 22 signals and the tracking device provides 6 signals for each hand. The authors reported that the fusion at the decision level outperforms the traditional feature-based fusion. They started with some basic experiments using the glove-based and the electromagnetic-based tracking systems independently. The

tracking system achieved a lower accuracy of 84.7% while the glove-based system achieved a better accuracy of 91.3%. The authors claimed that the traditional feature based combination provided a maximum accuracy of 96.2% which was improved to 98.1% when fusion at the decision level is performed. They built a database of 100 signs performed by a volunteer from the deaf community.

Other motion sensing devices exist that use infrared, radio frequency, ultrasound, magnetic and vibration based sensors. In mid 2012, the first leap product was publicly announced as a touchless sensor that can track hand and finger motion and provide effective descriptor of hand skeleton dynamics (De Smedt et al., 2016; Shin and Kim, 2020). The Leap Motion Controller (LMC) (Weichert et al., 2013) uses three infrared (IR) projectors (Light Emitting Diodes or LEDs) and two high precision monochromatic IR cameras to capture information within its operating range. The device detects and tracks the hand, finger and finger-like objects' positions, gestures, and motion. In late 2013, Leap Motion partnered with Asus and HP to embed the sensor in some keyboards and laptops. Since 2019, the peripheral sensor has been sold under the brand name Ultraleap from UltraHaptics. A similar technology is also employed in Microsoft Kinect using IR projector and camera for creating a depth image.

The sensor-based devices have the advantage of reducing the need for computer vision techniques to detect and segment the signer body. This enables the recognition system to recognize the sign gestures with low processing power. In addition, these devices can track the signer's motion that help in getting spatial and temporal information about the performed signs. However, some of these devices (e.g. gloves) require the signer to wear the sensor device during signing which limits its usage for real time applications. In addition, other sensing devices (e.g. LMC) cannot capture the overlapped signer's hands and they have the issue of low operating range that requires the signer to stand closely to the sensing device.

3.2. Vision-based acquisition techniques

In vision-based acquisition, video cameras are used to capture the sign gestures and store them as sequence of images for processing (Sharma and Singh, 2020). A major potion of reviewed works uses a single RGB video camera making it possible to adopt webcam or mobile phone cameras as an input device. RGB camera is also used in Microsoft Kinect in addition to the IR camera. Kinect sensor has three kinds of output, which can be used separately or in combination: color image, depth image, and skeleton image. The depth image is produced by CMOS IR sensor and can be used to create a mask image to remove the background by simple thresholding. Kinect can track skeletons with 18 joints (Kinect V1) or 25 joints (Kinect V2).

Multi-camera and 3D systems may help in lessening some of the environmental restrictions, but causing additional computational load which can be handled by the rapid advancement of computing systems. For instance, Tolba et al. (2012b) used two synchronized cameras positioned horizontally aligned with 90 degrees between them. Signer's hands should be located in a specific area with equal perpendicular distance from both cameras otherwise the sign would not be captured. The cameras capture 19 consecutive pair of images each pair is 5 degrees rotated from the previous pair. The ten odd-ordered pairs are chosen for training and the other 9 pairs for testing. These images are fed to Pulse-coupled Neural Networks (PCNN) to extract 2D signatures which are then weighted and combined linearly to produce 3D features.

Comparing with sensor-based, vision-based techniques are more applicable for SLR systems. They utilize video camera and do not require any spacial sensors. In addition, the vision-based devices do not have the issue of low operating range of sensor-based devices. They also do not require attaching the sensors to the signer body which make it appropriate for real-time applications. However, these devices provide a raw video stream that may need further preprocessing to make it convenient for feature extraction such as signer detection, background removal, and motion tracking. Moreover, it is required to handle the high degree of variability and sources of errors that are common in computer vision systems. These issues include noise and environmental factors due to variations in illumination, viewpoint, cluttered background, hand shape, posture of body parts, orientation, scale, and occlusion (Ong and Ranganath, 2005; Steger, 2002; Itti, 2002). However, the degree of severity of these issues varies in a practical implementation of a sign language system and is not always high due to working in controlled environment.

4. Sign language datasets

One of the main challenges of sign language recognition and translation systems is the availability of databases (Luqman and Mahmoud, 2020). Few databases are available for sign languages compared with gesture databases. Although sign language recognition is one of the applications of gesture recognition, gesture databases are rarely used for sign language recognition for several reasons. First, the number of classes in gesture recognition databases is usually limited. Secondly, sign language employs manual and non-manual gestures simultaneously, which is usually difficult to find them both annotated in one gesture database. In addition, sign language depends mainly on hand gestures while gesture databases are more broad and contain gestures related to the full body movements. Moreover, hand and finger details that should be encoded for sign language recognition systems are usually unavailable with gesture databases.

Several factors are important for evaluating sign language databases. One of these factors is the number of signers involved in performing the database signs. This factor adds more variance to the database and helps in evaluating the generalization of the recognition systems. The number of signs of the database is also important especially with isolated and continuous databases. Another factor is the number of samples per sign which is an important for training systems that require several samples of the same sign such as machine learning techniques. In addition, annotation of the continuous databases by providing the temporal information of continuous sentence components is also important for this type of databases.

Table 1 lists the most common databases used for sign language recognition categorized by the sign level (viz. fingerspelling, isolated, and continuous). The databases listed in this table are databases dedicated for sign language recognition and made available for researchers. As depicted in the table, the majority of the available sign language databases are isolated signs database. The number of signers in each database is important for training and evaluating recognition systems. Increasing the number of signers serves signer-independent recognition systems that train on signs performed by signers different from those involved in the evaluation. Although RGB video format is used for signs of all isolated signs databases, some databases provided depth and skeleton information of that signs as a result of using multi-modal devices such as Kinect.

Fingerspelling databases target only alphabets and/or digits of sign languages. Samples of signs are shown in Fig. 6. Some databases only have the signs of the 10 digits. Databases that contain alphabets sometimes exclude alphabets that require motion such as 'j' and 'z' letters of ASL (Pugeault and Bowden, 2011). Fingerspelling signs can be learned and performed easily by non-expert signers, which makes the collection of these signs much easier than other databases. Moreover, the variance between signers who perform fingerspelling signs does not have a significant impact on the performance of recognition systems since most of the fingerspelling database images show only the signer hands. The majority of these databases are available as images since the captured signs do not contain motion (Pugeault and Bowden, 2011; Barczak et al., 2011; Ronchetti et al., 2016; Latif et al., 2019; Koller et al., 2016; Hosoe et al., 2017; Pisharady et al., 2014, 2013). In addition, no special camera hardware is required and a typical video camera is commonly used to collect signs of this database. However,

Table 1
Publicly available Sign Language recognition datasets.

Type	Year	Dataset	Language[a]	Signs/Vocabulary	Signers	Samples/Videos
Fingerspelling	2006	RTWH Fingerspelling (Dreuw et al., 2006)	DGS	35	20	1400
	2010	NUS hand posture dataset I (Pisharady et al., 2014)	Singaporean	10		240
	2011	ASL Fingerspelling A (Pugeault and Bowden, 2011)	ASL	24	5	
	2011	MNIST (Pugeault and Bowden, 2011)	ASL	24		34,627
	2011	(Barczak et al., 2011)	ASL	36	5	2,425
	2013	NUS hand posture dataset II (Pisharady et al., 2013)	Singaporean	10	40	2000
	2014	(Dahmani and Larabi, 2014)	ArSL (uniform)	30	24	2880
	2014	(Dahmani and Larabi, 2014)	ArSL (complex)	30	8	960
	2014	RWTH-PHOENIX-Weather 2014 Handshapes (Koller et al., 2016)	DGS	60		3359
	2015	PSL Fingerspelling ToF (Kapuscinski et al., 2015)	PSL	16	3	960
	2016	LSA16 (Ronchetti et al., 2016)	Argentine	16	10	800
	2017	(Oliveira et al., 2017)	ISL	26	6	468
	2017	(Hosoe et al., 2017)	Japan	41	10	8055
	2019	(Latif et al., 2019)	ArSL	32		54,049
Isolated	2002	Purdue ASL (Martínez et al., 2002)	ASL	184	14	3,576
	2005	RWTH-BOSTON50 (Zahedi et al., 2005)	ASL	50	3	483
	2010	IIITA-ROBITA (Nandy et al., 2010)	Indian	22		605
	2012	ASLLVD (Neidle et al., 2012)	ASL	3,300+	6	9,800
	2012	MSR Gesture 3D (Kurakin et al., 2012)	ASL	12	10	336
	2012	DGS Kinect 40 (Ong et al., 2012)	DGS	40	15	3000
	2012	GSL 982 (Ong et al., 2012)	Greek	982	1	4910
	2013	PSL Kinect 30 (Oszust and Wysocki, 2013)	PSL	30	1	300
	2015	PSL ToF 84 (Kapuscinski et al., 2015)	PSL	84	1	1,680
	2015	DEVISIGN-L (Chai et al., 2015)	CSL	2000	8	24000
	2016	LSA64 (Ronchetti et al., 2016)	Argentine	64	10	3200
	2016	Indian Kinect (Ansari and Harit, 2016)	Indian	140	18	5041
	2016	BosphorusSign Kinect (Camgöz et al., 2016)	Turkish	855	10	
	2019	BVCSL3D (Ravi et al., 2019)	Indian	200	10	20,000
	2020	BosphorusSign22k (Özdemir et al., 2020)	Turkish	744	6	22,542
	2020	ASL-100-RGBD DATASET (Hassan et al., 2020)	ASL	100	22	4,150
	2020	KSL DATASET (Yang et al., 2020)	KSL	77	20	1,229
	2020	WLASL100 (Li et al., 2020)	ASL	100	97	2,038
	2020	WLASL300 (Li et al., 2020)	ASL	300	109	5,117
	2020	WLASL1000 (Li et al., 2020)	ASL	1,000	116	13,168
	2020	WLASL2000 (Li et al., 2020)	ASL	2,000	119	21,083
	2021	KArSL (Sidig et al., 2021)	ArSL	502	3	75,300
Continuous	2007	RWTH-BOSTON-104 (Dreuw et al., 2007, 2008)	ASL	103 + 65	3	161 + 40
	2008	RWTH-BOSTON-400 (Dreuw et al., 2008)	ASL	483+74+36	4	633+106+104
	2008	SIGNUM (Von Agris et al., 2008b)	DGL	450	25	780
	2012	RWTH-PHOENIX-Weather (Forster et al., 2012)	DGL	911	7	1,980
	2014	RWTH-PHOENIX-Weather (Koller et al., 2016)	DGL	1,080	9	6,861
	2015	SignsWorld Atlas MS and NMS (Shohieb et al., 2015)	ArSL	264+	10	535+
	2018	MS-ASL100 (Joze and Koller, 2018)	ASL	100	189	5,736
	2018	MS-ASL200 (Joze and Koller, 2018)	ASL	200	196	9,719
	2018	MS-ASL500 (Joze and Koller, 2018)	ASL	500	2222	17,823
	2018	MS-ASL1000 (Joze and Koller, 2018)	ASL	1,000	222	25,513
	2020	TheRuSLan Kinect (Kagirov et al., 2020)	RSL	164	13	

[a]ASL: American Sign language, ArSL: Arabic Sign Language, CSL: Chinese Sign Language, DGS: German Sign Language (aka Deutsche Gebärdensprache or DGS), KSL: Korean Sign Language, PSL: Polish Sign Language (aka Polski Język Migowy or PJM), RSL: Russian Sign Language; NS: Manual Signs, NMS: Non-Manual Signs.

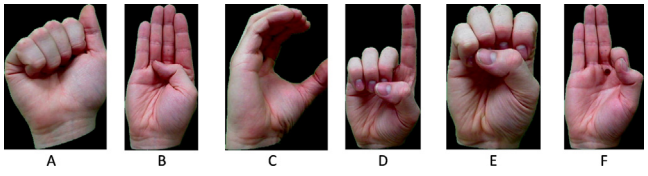


Fig. 6. Sample signs from fingerspelling database (Barczak et al., 2011).

some of these signs have similar hand gestures in some languages. For instance, in ArSL, 'Tah' is very similar to 'Zah', 'Rãa' is very similar to 'Zãy' letter, and 'Dãl' is very similar to 'Zãl'. Similarly in ASL, the signs for 's' and 't' letters are difficult to differentiate.

Isolated signs databases are the most common type of sign language databases. These databases contain isolated sign words performed by one or more signers. In contrast to fingerspelling databases, signs of these databases are motion-based signs which require more training for non-expert signers. Some of these databases may contain static signs as well and present them using video streams rather than images as fingerspelling databases (Ansari and Harit, 2016). In addition, variance between signers of isolated signs databases adds more challenge to the sign language recognition systems. Moreover, some signs have similar motion trajectories such as 'want' and 'what' in ASL. The most common acquisition technique used to capture the signs of these databases is vision-based techniques such as video camera. However, other acquisition devices that output more than one data are used recently for collecting sign words such as Kinect. More information about sign acquisition devices is available in Section 3.

Continuous sign language databases are composed of a set of sign language sentences where each sentence is a sequence of signs performed continuously. This type is more challenging than the former types and the number of available databases is relatively limited due to the difficulties associated with collection and annotation, which requires expert signers and more time for annotation. Besides the challenges for other types, it requires more effective segmentation methods or restricts signers to pause between different words. Some continuous databases, such as Shohieb et al. (2015), also include annotations of fingerspelling, isolated as well non-manual gestures such as facial expressions and lip motions. The acquisition devices used with these database range from video cameras to multi-modal devices such as Kinect.

Table 2
Summary of the surveyed fingerspelling recognition systems (Performance refers to accuracy unless indicated otherwise).

Paper	Year	Language	Methods	Signs/Signers	Performance
Munib et al. (2007)	2007	ASL	ANN + HT	20	92.3
Pugeault and Bowden (2011)	2011	ASL	RF + GF(4,4) + MKDI	24	Pr 75.0[b]
Wang et al. (2014)	2014	ASL	SP-EMD + MKDI	24	75.8
Aowal et al. (2014)	2014	ASL	2D-ZM	10	98.5
Dahmani and Larabi (2014)	2014	ArSL (UBG)	SVM + Skin	30/24	88.87[b]
Dahmani and Larabi (2014)	2014	ArSL (CBG)	SVM + Skin	30/8	86.9[b]
Ranga et al. (2018)	2018	ASL	CNN + (DWT, GF)	24	97.01[a], 76.3[b]
Bheda and Radpour (2017)	2017	ASL	CNN	26	82.5
Zamani and Kanan (2014)	2014	ASL	Salient detection + PCA + LDA	36	99.8
Pan et al. (2016)	2016	ASL	Hu Moments + BOW + FDs	36	94.0
Pan et al. (2016)	2016	CSL	Hu Moments + BOW + FDs	26	99.8
Chansri and Srinonchat (2016)	2016	Thai	ANN+ HoG + MKDI + Color	24	84.0
Anand et al. (2016)	2016	Indian	NN + DWT	13	99.23
Pattanaworapan et al. (2016)	2016	ASL	ANN + DWT	23/50	89.38[b]
Ezel and Baykan (2017)	2017	Turkish	CNN + SVM	–	84.5
Ameen and Vadera (2017)	2017	ASL	ConvNet + MKDI + Color	24	Pr 82.0[b]
Chevtchenko et al. (2018)	2018	ASL	ZM + Gabor	36	97.6
Tao et al. (2018)	2018	ASL	CNN + multiview + MKDI	26	–
Islam et al. (2018)	2018	ASL	CNN + SVM	10	94.6
Makhashen et al. (2019)	2019	ASL	Gabor + SVM	24	95.0
Aly et al. (2019)	2019	ASL	SVM + PCANet + MKDI	24	88.7[b]

[a]Signer-dependent.

[b]Signer-independent.

Pr = precision, HT: Hough Transform, ANN: Artificial Neural Network, DWT: Discrete Wavelet Transform, GF: Gabor Filter, CNN: Convolutional Neural Network, MKDI: Microsoft Kinect Depth Images, RF: Random Forest, SP-EMD: Superpixel Earth Mover's Distance; HoG: Histogram of Oriented Gradients, NN: Nearest Neighbor, UBG: Uniform Background, CBG: Complex Background.

5. Sign language recognition techniques

Several techniques have been proposed during the last decade for sign language recognition. Some of these techniques target static sign gestures including alphabet and digit signs while other techniques target dynamic sign gestures. In this section, we will classify the surveyed techniques into three categories fingerspelling, isolated, and continuous sign gestures recognition systems.

5.1. Fingerspelled signs recognition systems

A number of recognition techniques have been proposed in the literature with focus on the fingerspelled signs using traditional and deep learning approaches. Table 2 summarizes various example fingerspelling recognition systems.

Munib et al. (2007) proposed a system to recognize 20 alphabets of ASL using Hough transform for features extraction and neural network for classification. Using a dataset consisting of 300 images performed by 15 signers, an accuracy of 92.3% is reported. More recently, Ranga et al. (2018) used Convolutional Neural Network (CNN) to recognize 24 gestures of ASL database (Barczak et al., 2011). This database is divided into training and testing in two different ways signer-dependent and signer-independent. The signer-dependent dataset has all signers' images shuffled and divided into training and testing sets whereas the signer-independent dataset uses four signers for training and the fifth unseen signer for testing. Using these two datasets, the best reported accuracies were 97.01% and 76.25% on signer-dependent and signer-independent datasets, respectively. Using 26 alphabets of the same database, an accuracy of 82.5% was achieved by Bheda and Radpour (2017) by employing another model of CNN.

A saliency detection technique is used by Zamani and Kanan (2014) for recognizing fingerspelled signs of ASL. The proposed technique locates the important parts, called salient parts, in the image. These parts are fed into principal component analysis (PCA) and linear discriminant analysis (LDA) and the output is classified using neural network classifier. This approach achieved an accuracy of 99.88% on a dataset consisting of 36 gestures (Barczak et al., 2011). A combination of PCA, LDA, and Support Vector Machine (SVM) are used by Pan et al. (2016) to classify features extracted using Hu Moments, Bag-of-Words (BOW), and Fourier Descriptors (FDs). This hierarchical classification with the combined features achieved accuracies of 94% and 99.8% on ASL and CSL datasets, respectively. Zernike moments (ZMs) were also used by several researchers for gesture recognition and the reported results show that they were better than Hu Moments, PCA, and FDs (Aowal et al., 2014; Sabhara et al., 2013; Otiniano-Rodrıguez et al., 2012). In Aowal et al. (2014), 2D-ZMs outperformed PCA and FDs on classifying 10 alphabets of ASL with an accuracy of 98.5%. ZMs were also used by Chevtchenko et al. (2018) to recognize 36 gestures of ASL. These features were concatenated with Gabor features and achieved an accuracy of 97.6%.

Tao et al. (2018) used a multi-view augmentation fusion with CNN for ASL alphabets recognition using the depth images. Several views are generated from different perspectives to address the issues of occlusion and perspective variations. The predictions of these views are fused to form the final sign classification. CNN is also used by Ezel and Baykan (2017) for recognizing letters of Turkish SL and an accuracy of 84.48% is reported using 3-fold cross validation technique on a dataset consisting of 522 pictures. Islam et al. (2018) used CNN for extracting features from ASL digits whereas SVM is used for classification to achieve 94.6% accuracy. Texture based features were extracted by Makhashen et al. (2019) through down-sampling Gabor-transformed images. This approach achieved more than 95% accuracy on ASL alphabets.

A Bayesian model of visual attention is used by Pisharady et al. (2013) for hands segmentation. Hand shape and texture features were extracted from the segmented hands and fed into SVM for classification to achieve an accuracy of 94.36% using 2,750 hand postures. A hybrid feature fusion of Hu moment invariant, finger angle count, and angles of skin and non-skin colors were used by Yun et al. (2012). To evaluate this approach, a dataset consisting of 10 gestures is used and an accuracy of 91% was reported. Other researchers used Local Binary Pattern (LBP) texture operator for fingerspelling recognition of several sign languages including American (Ding et al., 2011), Arabic (Sidig et al., 2017), Bangladeshi and Chinese (Jasim and Hasanuzzaman, 2014) Sign languages.

5.2. Isolated signs recognition

Several techniques have been proposed for isolated signs recognition. Table 3 lists a number of proposed approaches. The majority

Table 3
Summary of recognition systems for isolated signs.

Paper	Year	Language	Sign language components		Signs/Signers	Accuracy
			Manual	Non-manual		
Gao et al. (2004a)	2004	CSL	✓		5113/6	82.9[b]
Shanableh et al. (2007)	2007	ArSL	✓		23	87.0
Aran et al. (2007b)	2007	ASL	✓	✓	8	81.6
Paulraj et al. (2008)	2008	Malaysian	✓	✓	32	92.1
Von Agris et al. (2008b)	2008	DGS	✓	✓	450	80.2
Al-Rousan et al. (2009)	2009	ArSL	✓		30/18	96.7[a]/94.2[b]
Zaki and Shaheen (2011)	2011	ASL	✓		50	89.1
Mohandes et al. (2012)	2012	ArSL	✓		300	95.0
Prasuhn et al. (2014)	2014	ASL	✓		26	47.0
Almeida et al. (2014)	2014	Brazilian	✓		34	80.0
Huang et al. (2015)	2015	ASL	✓		25	94.2
Li et al. (2015)	2015	ASL	✓		24	99.1
Fathy et al. (2015)	2015	ArSL	✓	✓	26	90.0
Pan et al. (2016)	2016	ASL	✓		26	94.0
Pan et al. (2016)	2016	CSL	✓		26	99.8
Jin et al. (2016)	2016	ASL	✓		16	97.1
Pattanaworapan et al. (2016)	2016	ASL	✓		23	89.4
Thalange and Dixit (2016)	2016	ASL	✓		10	98.2
Liu et al. (2016)	2016	CSL	✓		500	63.3
Hartanto and Kartikasari (2016)	2016	Indonesian	✓		24	91.7
Quesada et al. (2016)	2016	ASL	✓	✓	9	91.3
Nai et al. (2017)	2017	ASL	✓		10	89.6
Kumar and Manjunatha (2017)	2017	ASL	✓		10	99.0
Sidig and Mahmoud (2018)	2018	ArSL	✓		100/3	99.7[a]/64.0[b]
Kumar et al. (2018)	2018	ISL	✓	✓	51	94.3
Sabyrov et al. (2019)	2019	Kazakh-Russian	✓	✓	20	73.0
Ravi et al. (2019)	2019	ISL	✓		200/10	86.66[a]/74.58[b]
Lim et al. (2019)	2019	ASL	✓		15/3	89.33
Wadhawan and Kumar (2020)	2020	ISL	✓		100/-	99.72C, 99.9G[c]
Aly and Aly (2020)	2020	ArSL	✓		23/3	89.59[b]
Azar and Seyedarabi (2020)	2020	Persian	✓		20/12	97.6[a]/96.7[b]

[a]Signer-dependent.
[b]Signer-independent.
[c]C: Colored, G: Grayscale.

of these techniques target either manual (e.g., hand gestures) or non-manual (e.g., facial expressions and lip motions) components of the sign language. However, few approaches studied the combination of these two components for boosting the recognition accuracy of computer-based sign language systems. This section reviews various approaches for isolated signs recognition. Manual-based approaches employ only manual gestures for sign language recognition whereas hybrid-based approaches integrate manual and non-manual gestures. Isolated sign recognition is also important for techniques that convert continuous sign language recognition into isolated signs.

5.2.1. Manual-based sign recognition approaches

Manual-based approaches depend mainly on hand gestures for sign language recognition. These approaches can be classified into approaches that use classical features extracted from signer hands (Nai et al., 2017; Almeida et al., 2014; Tolba et al., 2012a; Ng and Ranganath, 2002; Pattanaworapan et al., 2016; Thalange and Dixit, 2016; Aowal et al., 2014; Sabhara et al., 2013; Otiniano-Rodrıguez et al., 2012; Shanableh et al., 2007; Zaki and Shaheen, 2011; Pan et al., 2016; Hartanto and Kartikasari, 2016; Celebi et al., 2013; Plouffe and Cretu, 2015; Jin et al., 2016; Kumar and Manjunatha, 2017; Mohandes et al., 2012) and approaches that employ machine learning techniques, such as deep learning, for sign recognition (Cui et al., 2017; Huang et al., 2015; Li et al., 2015; Barros et al., 2014).

A) Using classical features

A Gaussian skin color model is employed by Mohandes and Deriche (2005) to detect the signer face that is used later as a reference for hands tracking. Then, a Hidden Markov Model (HMM) was used to classify a set of geometric features extracted from the detected hand regions to achieve an accuracy of 92% using 50 signs. This approach depends on colored gloves worn by the signer to help in hands tracking and segmentation. In Mohandes et al. (2012), the authors investigated the number of HMM states that give the best accuracy on 300 signs of ArSL. The signer face was detected using a Gaussian skin color model and the face centre was used as a reference point to track the hand motion. A set of statistical features were extracted from the segmented hand and fed into HMM. The authors reported that the best accuracy with about 95% is obtained using six states of HMM. However, this technique requires the signer to wear colored gloves during signing.

Nai et al. (2017) proposed a system to recognize static gestures using information extracted from depth images. A set of statistical features were extracted from the hand image and an accuracy of 89.6% was reported using ASL digits. Almeida et al. (2014) extracted seven features related to the structural elements from depth images of Brazilian Sign Language and an accuracy above 80% was reported on 34 signs. Zaki and Shaheen (2011) extracted joint features using motion chain code (MCC), Kurtosis position, and Principle Component Analysis (PCA) to represent the hand shape and orientation of ASL gestures. Kurtosis position was used to detect hand articulation position and MCC was used to encode the hand motion direction. Each extracted feature was fed into HMM and the final classification was used based on majority voting.

A combination of PCA, LDA, and SVM were used by Pan et al. (2016) for sign language recognition. Signer hands were segmented using an adaptive skin color segmentation algorithm which depends on the user to move his hand in front of the camera to adjust the skin color range. Then, Scale-Invariant Feature Transform (SIFT), HU Moments, and Fourier transform descriptors were used for features extraction and accuracies of 99.8% and 94% were reported on Chinese and American sign languages, respectively. However, depending on the user to adjust the skin color range is a limitation of this technique. Hands segmentation based on skin color was also used by Hartanto and Kartikasari (2016) for segmenting 24 signs of Indonesian Sign Language. Dynamic-Time Warping (DTW) is used by Celebi et al. (2013) and Plouffe and Cretu (2015) for static and dynamic gestures recognition.

Jin et al. (2016) used a Canny detector followed by seeded region growing for hand detection and segmentation. A Bag of Features (BoF) technique was used with features extracted using Speeded Up Robust Features (SURF) algorithm. Then, SVM was used to classify 16 gestures of ASL and an accuracy of 97.13% was reported. Histogram of Gradients (HOG) was used by Prasuhn et al. (2014) to extract features from a hand area that is segmented by hand binarization in the Hue-Saturation-Value (HSV) color space. Kumar and Manjunatha (2017) compared Superpixels, LBP, and SURF for ASL recognition and reported that SURF with a K-Nearest Neighbor (KNN) classifier obtained the highest accuracy compared with other techniques.

Tolba et al. (2012a) used two cameras positioned horizontally and aligned with 90 degrees between them to capture ArSL signs. Each sign gesture was treated as a sequence of postures and Pulse-coupled Neural Networks (PCNN) was used to extract the features. The accuracy of this approach dropped sharply when the signer percentage was under 40% of the frontal view because this approach depends on the position of the signer with respect to the camera, which is a limitation. Ng and Ranganath (2002) used Fourier descriptors to extract features from hand gestures and Radial-Basis Function (RBF) network was used for pose classification. The pose vector outputted from the RBF network was inputted into an HMM and a Recurrent Neural Network (RNN) and an accuracy of 91.9% was reported using a linear combination of HMM and RNN. Pattanaworapan et al. (2016) used Discrete Wavelet Transform (DWT) and Area Level Run Lengths to extract features from ASL gestures. DWT was used to extract the features from signs without finger extended out of the palm while Area Level Run Lengths were used with signs with at least one finger extended. The authors reported accuracies of 89.38% and 27.86% for non-fist and fist signs, respectively. Wavelet and combined orientation histogram with statistical features were used by Thalange and Dixit (2016) for recognizing ASL digits and the best reported accuracy of 98.17% was using wavelet features. Azar and Seyedarabi (2020) extracted hands trajectory from signer hands worn colored gloves using region-growing technique. The trajectories along with other hand shape information were modeled using HMM. This technique was evaluated using a dataset consisting of 20 signs of Persian SL performed by 12 signers and accuracies of 97.63% and 96.70% were reported with signer dependent and independent strategies, respectively. Trajectory is also used by Sidig and Mahmoud (2018) with 100 signs of KArSL database and they reported 99.7% and 64.0% accuracies with signer dependent and independent settings, respectively.

Shanableh et al. (2007) used temporal and spatial features for recognizing 23 signs of ArSL. For temporal features, each sign video was summarized into one image using forward accumulated difference of successive frames. Two approaches were used to extract the spatial features from the accumulated image. The first approach used DCT followed by Zigzag-Zonal coding, while the second approach used Radon transformation followed by low-pass filtering. Signer-dependent and signer-independent experiments were conducted. For signer-dependent, all signers were involved in the training and testing of the system and an accuracy of 93.84% was reported while for signer-independent, tests were on signer that was not seen before and an accuracy of 87% was reported. However, this approach is sensitive to the environment conditions, such as background color, texture, and lighting conditions, that may result in performance degradation (Zhang et al., 2011).

B) Using deep learning

Cui et al. (2017) employed CNN and RNN with Long short-Term Memory (LSTM) for recognizing continuous sign language sentences. The proposed technique was evaluated using RWTH-PHOENIX-Weather 2014 dataset (Koller et al., 2015a) and an accuracy of 61.3% was reported. LSTM was also used by Liu et al. (2016) for recognizing sign language gestures captured using Kinect sensor camera. A feature vector of 12 features, representing the skeleton joints of left and right hands, were used as input the model that consists of seven layers. Two datasets consisting of 25 K and 125 K images were used to evaluate their work and accuracies of 86% and 64% were achieved, respectively. Spatial–temporal features extracted from Kinect video were used by Huang et al. (2015) as input for a CNN model consisting of eight layers. A multi-channel input consisting of color, depth, and skeleton information was used as input to this model and an accuracy of 94.2% is reported on a dataset of 25 signs developed by the authors.

A combination of color and depth images were used by Li et al. (2015) to recognize ASL. Sparse auto-encoder (SAE) with CNN was used to learn features from color and depth images. The learned features were concatenated and fed into PCA to get the final feature vectors which were classified using a linear SVM. This technique was evaluated using 20 K images representing 24 alphabets of ASL (Pugeault and Bowden, 2011) and an accuracy of 99.05% was reported. A multi-modal CNN was also used by Neverova et al. (2014) for classifying dynamic gestures by concatenating color, depth, and pose information. Barros et al. (2014) used CNN with input of three channel images. The first channel is the grayscale image of the hand gesture while the other two channels are the vertical and horizontal gradients of the same image extracted using Sobel operator. The authors evaluated their technique on a database of 10 signs and an F1-score of 94% was reported.

5.2.2. Hybrid-based sign recognition approaches

Few approaches studied the combination of manual and non-manual components of sign language for boosting the accuracy of computer-based SLR systems. Theses approaches utilized vision-based acquisition devices to capture manual and non-manual features since the non-manual features could not be captured by sensor-based devices such as gloves and LMC. Sabyrov et al. (2019) studied the importance of different non-manual components for improving the accuracy of Kazakh-Russian SL recognition systems. The authors used Logistic Regression (LR) to classify key points that were extracted from facial expressions and manual gestures of 20 signs using OpenPose library. The results of this study showed that combining only eyebrows key points with manual features improved the accuracy by only 0.5% whereas mouth key points improved the accuracy by 7%. Z. et al. (2010) studied the correlation between manual and non-manual components. The obtained results on Czech SL showed that the hand and head gestures are correlated mainly in signs with vertical movement of head and hands. Elons et al. (2014) proposed another system using recursive principal components and multilayer perceptron and found that the addition of facial features to hand sign raised the recognition rate from 88% to 98%.

Aran et al. (2007b) proposed a sequential two-step method for combining manual and non-manual features extracted from 8 signs of ASL. The first stage used HMM for classifying signs using both features and the second stage was used only to handle uncertainty in classification of the first stage. This latter stage employs only non-manual features using likelihood calculation inside the sign cluster. This approach improved the recognition rate by 14.5% compared with using only manual features. Paulraj et al. (2008) segmented the signer head and hands regions from a sequence of image frames representing sign video. Then, a feature vector consisting of the area and 15 DCT coefficients were extracted from each segmented region. The feature vectors of all regions were combined and fed into a simple neural network model to classify 32 signs of Malaysian Sign Language and an accuracy of 92.1% was reported. This technique requires the signer to wear predetermined colored clothes to facilitate the detection and segmentation of the signer head and hands which makes this approach unsuitable for real-time recognition systems.

Kelly et al. (2009a) proposed a sign language recognition system by incorporating a set of statistical features extracted from the signer's head and hands. The extracted features were fed into a multichannel HMM and a detection ratio of 95.7% was achieved on a dataset consisting of 160 video clips of unsegmented sentences with a small vocabulary of eight manual signs and three head movement gestures.

The system requires the signer to wear colored gloves as cues for segmenting hands and tracking head motion, which forms a limitation of this technique. Rao and Kishore (2018) employed a 2D-DCT for Indian Sign Language recognition. To detect the hand and head positions, a Sobel edge detector was used after applying Gaussian filter on each video frame. Then, a set of features were extracted from hand and head contours using DCT. A dataset of 18 sign videos recorded using the front-facing mobile camera was used to evaluate the proposed work and an accuracy of 90.58% was attained. DCT is also used by Al-Rousan et al. (2009) for ArSL recognition. DCT was followed by Zonal coding and fed into HMM for classification. This technique was evaluated on a dataset consisting of 30 signs performed by 18 signers using signer dependent and independent modes. 96.74% accuracy was reported for signer dependent and 94.2% was reported for signer independent.

The second class of techniques that integrated non-manual features with manual gestures used two or more non-manual features with manual gestures. Von Agris et al. (2008b) employed both facial expressions and manual gestures for automatic recognition of GSL. Skin color was used to detect the face and hand of the signer. For facial expressions recognition, an active appearance was applied to recognize eyes and mouth regions. Then a numerical description was computed from those regions. For hands region, spatial features were extracted such as hand center coordinates relative to the shoulder position. Combining facial expressions with hand gestures improved the recognition accuracies of isolated and continuous signs by around 1.5% and 6%, respectively.

Sarkar et al. (2010) used skin color to detect and segment the hands and face of the signer. Then, the statistical distribution captured from the low-level primitives of the body parts was used to represent manual gestures as points in the Space of Probability Functions. Facial expressions were represented in another space using PCA, while head motion was represented as 2D image tracks. Combining manual and non-manual features improved the accuracy from 88% to 92% on a dataset consisting of 25 sentences with 39 distinct ASL signs. Caridakis et al. (2014) discussed the syntactic and grammatical relation of facial expressions to sign language. In addition, the efficiency of including facial expressions in sign language recognition was investigated.

Yang and Lee (2011) fused features extracted from signer hands and face. Hierarchical conditional random field (CRF) and BoostMap embedding were used for hands discrimination while active appearance model with SVM were used for facial expressions. This technique was trained and evaluated on a database of ASL consisting of 98 sentences with a vocabulary size of 24 signs performed by signers wearing colored gloves. Fathy et al. (2015) detected and segmented the signer face and hands based on skin color. The segmented hand was divided into 4×4 blocks and the centroid of each block was used to build a feature vector of the hand. To recognize facial expressions, locations of eyes, nose, and mouth were extracted. This work evaluated hand gestures and facial expressions separately without integration. In addition, the dataset that was used to evaluate this work consists of ArSL alphabets containing static signs that do not depend mainly on facial expressions.

Quesada et al. (2016) used SVM classifier to classify hand shapes and non-manual markers that were described using Natural User Interface device of type Intel RealSense. This device uses 3D cameras to capture and recognize hand shapes, face expressions, and body position. Using a small ASL dataset consisting of 5 face gestures and 4 handshapes, an accuracy of 91.25% was reported. The authors in Kumar et al. (2018) used Leap Motion Controller and Kinect sensors to capture hand gestures and facial expressions, respectively. HMM was used for recognizing each modality which were then combined using a Bayesian classification method. The proposed method was evaluated on 51 sign words of Indian Sign Language and an improvement of 1.04% over single modality accuracy was achieved.

5.3. Continuous signs recognition

Continuous sentence of sign language incorporates more than one sign performed continuously without pausing between signs. Continuous SLR (CSLR) is more challenging endeavor than isolated sign recognition due to the multimodal nature of the visual cues (manual and non-manual gestures) (Piater et al., 2010). Two main problems are usually associated with CSLR: sign sequence segmentation and recognition (Yang et al., 2016; Aloysius and Geetha, 2020).

Continuous sign language sentences are naturally signed and movement epentheses between two sentences' signs are not similar and differ from one signer to another. In addition, the lack of an accurate temporal segmentation information for sentence signs makes finding an effective segmentation algorithm a critical issue of CSLR, in particular with large vocabulary size (Fang et al., 2006). Some systems convert the problem of CSLR into isolated SLR by first segmenting the sentence into signs or small segments and then perform recognition at the sign or segment levels. However, having an accurate segmentation system for continuous SL sentences is still an open research problem and we are not aware of the availability of such system. To overcome this issue, recent CSLR systems use techniques that do automatic labeling and recognizing of sentences without the need for prior segmentation. HMM and Connectionist Temporal Classification (CTC) are examples of the techniques that do not require pre-segmentation of the sentence and do automatic alignment between sentence segments and labels.

Liang and Ouhyoung (1998) used time-varying parameters (TVP) for segmenting continuous Taiwanese Sign Language (TWL) sentences by thresholding the number of TVPs of the hand flexion and consider it as the end of the posture. Khan et al. (2014) proposed an approach to segment continuous sentences of New Zealand Sign Language (NZSL) into basic lexical units. A delayed absolute difference (DAD) algorithm is employed for pauses detection by converting the continuous sentence into a set of segmentation features to be used later for searching for sign boundaries. According to the reported results, DAD-based pause detection algorithm outperformed three pause detection algorithms (velocity-based method, MH model, and the TVP).

A simple recurrent network (SRN) technique was proposed by Fang et al. (2001) to segment CSL sentences collected using two data gloves and three position trackers into signs. Each sentence was segmented according to divide-and-conquer approach and to find the best sequence of the segmented signs HMM was used with Lattice Viterbi algorithm. Gao et al. (2004b) proposed a system that matches input data with movement epentheses templates using DTW. However, this system assumes that movement epentheses between two signs are similar in different sentences which is invalid assumption especially with real-world sentences (Yang et al., 2008). DTW was also used with HMM by Zhang et al. (2014) for CSLR. DTW was employed with Length-Root method where DTW detects the endpoint of the sign while Length-Root method selects the optimal candidate endpoint. Kong and Ranganath (2014) segmented ASL continuous sentences into segments categorized as SIGN or movement epenthesis. Segmentation was performed by fusing the outputs of CRF and SVM classifiers using Bayesian network (BN). The sub-segments labeled as SIGN were concatenated and recognized using two-layer CRF classifier. Another idea for sentence segmentation was presented by Xu et al. (2018) using key frames. The key-frames were considered as the basic unit of sign word which can be organized into meaningful sentences. These key-frames were extracted with the help of Kinect using the trajectory of the sign and each key-frame was given a weight based on its semantic contribution.

CSLR systems can be broadly sub-divided into classical, HMM-based and deep learning based techniques. Most of the techniques proposed for CSLR before 2015 employed HMM for recognition and used sensor-based devices for data acquisition. In contrast, CSLR techniques after 2015 used deep learning for recognition and vision-based devices for data acquisition. Table 4 lists a summary of CSLR techniques which are organized into three groups and more details are provided in the following subsections.

Table 4
Continuous sign language recognition techniques.

Paper	Year	Language	Acquisition method		Sentences/Datasets	Performance	
			Sensor	Vision		Accuracy	WER
Liang and Ouhyoung (1998)	1998	TSL	✓		196	80.4	
Bauer and Hienz (2000)	2000	DGS		✓		91.7	
Wang et al. (2001)	2001	CSL	✓		100	90.0	
Fang et al. (2001)	2001	CSL	✓		100	85.0	
Yuan et al. (2002)	2002	CSL		✓	40	70.0	
Gao et al. (2004b)	2004	CSL	✓		1500	90.8	
Yao et al. (2006)	2006	CSL	✓		543	70.2	
Cortés et al. (2006)	2006	Spanish		✓	59	99.5	
Infantino et al. (2007)	2007	Italian		✓	80	82.5	
Von Agris et al. (2008a)	2008	DGS		✓	780	75.8	
Kelly et al. (2009a)	2009	ISL		✓	160	95.0	
Yu et al. (2011)	2011	TSL		✓	3	67.0	
Gweth et al. (2012)	2012	DGS		✓	SIGNUM	84.0	
Forster et al. (2013)	2013	DGS		✓	SIGNUM		
Forster et al. (2013)	2013	DGS		✓	RWTH-PHOENIX-Weather		
Zhang et al. (2014)	2014	CSL		✓	180	82.2	
Koller et al. (2014)	2014	DGS		✓	5	47.1	
Yang et al. (2016)	2016	CSL		✓	20		
Fang et al. (2017)	2017	CSL		✓	100		16
Ekiz et al. (2017)	2017	Turkish	✓		13	97.6	
Cui et al. (2017)	2017	DGS		✓	RWTH-PHOENIX-Weather		38.7
Camgoz et al. (2017)	2017	DGS		✓			40.7
Rao et al. (2017)	2017	Indian		✓	18	95.6	
Koller et al. (2018)	2018	DGS		✓	SIGNUM		7.4
Koller et al. (2018)	2018	DGS		✓	RWTH-PHOENIX-Weather		30.0
Pu et al. (2018)	2018	DGS		✓	RWTH-PHOENIX-Weather		37.3
Wei et al. (2019)	2019	CSL		✓	100		51
Zhou et al. (2019)	2019	CSL		✓	25K		4.7
Pu et al. (2019)	2019	CSL		✓	100		32.7
Elakkiya and Selvamani (2019)	2019	DGS		✓	RWTH-PHOENIX-Weather	91.0	
Cui et al. (2019)	2019	DGS		✓	RWTH-PHOENIX-Weather		22.9
Pu et al. (2019)	2019	DGS		✓	RWTH-PHOENIX-Weather		36.7
Pei et al. (2019)	2019	DGS		✓	RWTH-PHOENIX-Weather-2014		40.6
Zhou et al. (2019)	2019	DGS		✓	RWTH-PHOENIX-Weather-2014		34.5
Cui et al. (2019)	2019	DGS		✓	SIGNUM		2.8
Yang et al. (2019)	2019	DGS		✓	RWTH-PHOENIX-Weather-2014		34.9
Yang et al. (2019)	2019	CSL		✓	25K		3.8
Mittal et al. (2019)	2019	Indian		✓	942	72.3	
Suri and Gupta (2019)	2019	Indian	✓		20	94.0	
Elakkiya and Selvamani (2019)	2019	ASL		✓	ASLLVD	85.4	

5.3.1. Classical CSR techniques

Nayak et al. (2005) employed continuous state space model for recognizing continuous sentences of ASL. This technique is based on extracting and learning sign patterns called signemes from continuous ASL sentences using unsupervised approach. These signemes were used as templates to be matched with sub-parts of the testing sentences. However, these signemes are biased by the first two sentences used for extracting them (Nayak et al., 2009). The idea of extracting signemes was also used in Nayak et al. (2009) using Iterated Conditional Modes (ICM). This technique correctly extracted a set of 10 signemes from 96 sentences out of 136 ASL sentences. Infantino et al. (2007) proposed a framework for Italian SL by analyzing the correctness of a sentence. A skin detection algorithm was used to segment the signer's hands and head. Then, a set of geometrical features were extracted from each sign and classified using self-organizing map (SOM) neural network. This approach was evaluated using two datasets and accuracies of 83.3% and 82.5% were reported using 30 and 80 sentences, respectively.

Tripathi and Nandi (2015) proposed an approach to recognize 11 continuous sentences of Indian SL by first splitting the sentence into signs using gradient based key frame extraction method and performing the recognition at sign level. Orientation Histogram (OH) and DWT are used to extract features from the isolated signs and PCA is used for features reduction before applying distance measuring classifiers. In Sahoo et al. (2020), the authors applied soft computing techniques to Indian SL to convert acquired images or videos into words, numbers or sentences. Another idea for sentence recognition is presented in Xu et al. (2018) using key frames. The key frames of the sign words that constitute the sentence were extracted from training samples and used as templates for matching new sentences. Wang et al. (2016a) proposed a CSL recognition system using RGB and depth data collected using Kinect sensor. Hand postures and movements were represented using HOG and 3D hand features, respectively, and a template matching was used for recognition where both 3D hand trajectories and hand postures were matched against the pr-extracted sign templates.

Smart watch sensors were used for data collection by Ekiz et al. (2017). These sensors generate accelerometer and gyroscope data that are used as features and fed into LR classifier. This approach was evaluated on 8 questions and 13 sentences yielding accuracies between 85.7% and 97.6%. A custom designed wearable IMU device is used by Suri and Gupta (2019) for SL acquisition. This device consists of a set of sensors (accelerometer and gyroscope) that provide signals about the performed signs. The information obtained using these sensors were employed by one-dimensional deep capsule network for Indian SL recognition. This network uses Capsule theory and consists of nested convolutional layers linked to each other by dynamic routing. This approach was evaluated on 20 sentences and an accuracy of 94.0% was reported.

5.3.2. HMM-based CSLR techniques

The work of Liang and Ouhyoung (1998) is one of the oldest approaches proposed for CSLR. This approach used DataGloveTM gloves for collecting 196 sentences of TSL and HMM for recognition. Bauer and Hienz (2000) extracted a feature vector from the sign video to model manual sign parameters and used it as an input to HMM for classification. The features vector consists of the location, hand shape, and orientation of the signer fingers, hands and body. To detect signer

fingers and hands, colored gloves were worn by a single signer involved in performing 97 signs of DGS. These features were classified by HMM with one model for each sign and an accuracy of 91.7% was reported. This result was improved to 93.2% when bigram language model was employed in Bauer et al. (2000). Wang et al. (2001) utilized two CyberGloves and 3D tracker for acquiring 100 sentences of CSL. These sensors represented each sign through hand postures, position, and orientation which were fed to HMM resulting in over 90% accuracy.

Yuan et al. (2002) used HMM to recognize CSL at subunit level of the sign. Although the definition of subunits followed by authors is linguistically and is not a commonly agreed upon model (Ong, 2007), a recognition improvement by 5% was reported with subunits over sign level recognition. At sentence level, an accuracy of 70% was reported using 40 sentences of CSL where each sentence consisted of a maximum of 4 subunits. Fang et al. (2001) utilized two 18-senor data gloves and three position trackers for CSL acquisition. These sensors returned a data vector consisting of 48 features for both hands. Each sentence was segmented according to divide-and-conquer approach using SRN and to find the best sequence of the segmented signs, HMM was employed with Lattice Viterbi algorithm. The proposed system obtained 85% accuracy on a dataset consisting of 100 sentences performed by seven signers. Dynamic programming was also used by Gao et al. (2004b) with DTW for CSL recognition. This approach was evaluated on 1500 sentences collected using two Cybergloves and three position trackers and an accuracy of 90.8% was reported.

Cortés et al. (2006) used optical flow with HMM for recognizing continuous sentences of Spanish SL. Optical flow was used to build a features vector from sentences consisting of 33 signs recorded in a constrained environment where signer wore colored gloves to help in hands tracking and an accuracy of 99.5% was reported using HMM. Yao et al. (2006) used One-Pass pre-searching method before Viterbi searching to speed up the recognition process. The proposed technique recognizes a sentence by recognizing words that make up that sentence. One-Pass pre-searching method outputs a list of candidate words and Viterbi-Beam searching algorithm was used later to filter this list. This method was evaluated on 543 sentences performed by two signers wearing two Cybergloves and three position trackers. Although the recognition time was reduced sharply from 1.047 sec/word using Viterbi-Beam to 0.343 sec/word, the accuracy was dropped from 78.1% to 70.2% using the proposed method.

Von Agris et al. (2008a) proposed an adaptation approach consisting of eigenvoices, maximum likelihood linear regression, and maximum a posteriori estimation for signer-independent recognition of DGS. This approach used manual and non-manual gestures for sign language recognition. Manual features were extracted using several hand position tracking techniques while facial features are extracted using Active Appearance Model (AAM). These features were concatenated into one features vector and classified using HMM. This adaptation technique improved the accuracy to 75.8% compared to 65.3% without adaptation. Kelly et al. (2009a) employed also manual and non-manual gestures for CSLR. The position of hands, eyes, and face were extracted and used as input to multidimensional HMMs. This approach achieved a detection ratio of 0.95 on ISL sentences performed by signer wearing colored gloves. Kelly et al. in another work used only non-manual gestures for CSLR (Kelly et al., 2009b) whereas Koller et al. (2014) recognized non-manual gestures of DGS through lips reading and employed them for CSLR of DGS. Signer mouth was tracked using AAMs and ten distance measurements around the signer's mouth with the average of the color intensity of three areas inside the mouth were extracted to model 11 visemes. These features were clustered using Gaussian clustering and Expectation Maximization (EM) and used as input to HMM. This approach reached 47.1% precision on 5 annotated sentences of DGS.

Yu et al. (2011) used Product-HMMs (pHMM) to distinguish between signs and out-of-vocabulary signs of TSL. Colored gloves were worn by signers to help in hands detection and tracking. Then hands were tracked using CamShift algorithm and trajectory of hand movement was computed by calculating the center of gravity of hands region. The authors reported 67% accuracy on three sentences performed by three signers, which is low given the size of the dataset. A Gaussian HMM (GHMM) is used by Gweth et al. (2012) for automatic recognition of DGS. A combination of multilayer perceptron (MLP) based features and appearance-based features were extracted from sign images and used as input to GHMM. This technique was evaluated on 177 sentences of SIGNUM database and 84% accuracy was reported.

Four combination scenarios of five modalities of DGS were evaluated by Forster et al. (2013). They investigated early combination of features, late fusion of decisions, as well as combination on HMM state and gloss levels. The five explored modalities are facial expressions, shapes and orientations of right and left hands, full upper body and body pose, and motion trajectory and position. The proposed system was evaluated on SIGNUM and RWTH-PHOENIX-Weather datasets and the lowest WERs was obtained by combining hands and facial expression features. In addition, the reported results indicate that combinations on the HMM state level and on the gloss level outperformed other combinations. Zhang et al. (2014) employed the skeleton and depth information obtained from Kinect with discrete HMM (DHMM) for CSLR. The proposed approach consisted of two stages for segmentation and recognition. A sentence was segmented into signs using DTW and Length-Root method. DTW detects the endpoint of the sign while Length-Root method selects the optimal candidate endpoint. The corresponding features of the segmented signs were classified by DHMM and an accuracy of 82.2% was reported using 180 sentence samples performed by three signers. Yang et al. (2016) used HMM to compute the similarity between sign models and sign sequences of test data. To enhance the recognition, the sign length and grammar constraints were embedded into Level Building HMM (LB-HMM). To reduce the computation of LB-HMM, Fast-HMM was proposed to approximately compute the likelihood of LB-HMM. This algorithm outperformed LB-DWT and T-HMM using a dataset consisting of 20 sentences of CSL collected using Kinect.

Elakkiya and Selvamani (2019) employed Bayesian network to classify gesture subunits into spatial and temporal. These subunits were recombined using HMM and clustered using minimum entropy clustering and DTW. Subunit multi-stream parallel HMM (SMP-HMM) was then used to train the clustered subunits based on Bakis topology. This approach was evaluated on RWTH-PHOENIX-Weather and ASLLVD datasets and accuracies of 91% and 85.4% were reported, respectively.

5.3.3. Deep learning CSLR techniques

To benefit from the discriminative abilities of CNN with the sequence modeling capabilities of HMM in the context of SL recognition, several techniques have been recently proposed for CSLR by integrating CNN with HMM. Koller et al. (2016) presented a frame-based learning approach using weakly labeled sequence data. This approach proposed an EM-based algorithm by integrating CNN with HMM. In another work (Koller et al., 2018), the authors integrated CNN with HMM for DGS. This approach was evaluated in three datasets of DGS and improvements of over 15% were reported. Fang et al. (2017) incorporated hierarchical bidirectional deep recurrent neural network (HB-RNN) and CTC for sign-level and sentence-level recognition, respectively. Two datasets of 56 sign words and 100 ASL sentences were collected using Leap Motion sensor and WER of 5.5%, 8.2%, and 16.1% were reported for word-level, seen-signer sentences, and unseen-signer sentences, respectively. CNN-LSTM was also used by Cui et al. (2017) for DGS recognition. CNN was used for features extraction while LSTM was used for sequence learning. This method was evaluated on RWTH-PHOENIX-Weather database and a WER of 38.7% was reported.

In Rao et al. (2017), hands and head parts were segmented using Sobel edges detector followed by morphological and adaptive thresholding. Discrete Cosine Transform (DCT) was used to extract features from segmented parts while ANN was used for classification. This approach was evaluated using 18 signs performed by 10 signers recorded

using the frontal-camera of the mobile phone and an accuracy of 95.6% was achieved.

Camgoz et al. (2017) proposed a sequence-to-sequence model for CSLR called SubUNets. This model consisted of three models CNN, Bidirectional Long Short Term Memory Layers (BLSTM), and CTC. CNN was used for spatial modeling while BLSTM was used for temporal modeling. CTC was the final model in the SubUNet pipeline and it was employed for sequence-to-sequence modeling. This approach was evaluated on hand shape recognition and CSLR and achieved 80.3% accuracy and 40.7% WER, respectively. Pu et al. (2018) proposed a deep NN with iterative optimization strategy for CSLR. A 3D-ResNet was utilized to extract features and stacked dilated convolutional network with CTC was used for mapping between the extracted features and sentence labels. Iterative optimization was used to alleviate the problem of limited contribution of CTC loss to the early CNN parameters which makes the network hard to train. This approach was evaluated on RWTH-PHOENIX-Weather and a WER of 37.3% was reported.

Cui et al. (2019) employed CNN with stacked temporal fusion layers for features extraction and RNNs with Bi-LSTM for sequence learning. An iterative optimization technique was used to exploit the representation capability of the used networks with limited data. The sequence-to-sequence mode was first trained for alignment and then used to tune the features extraction module. In addition, fusing of optical flow with RGB images was also explored in this work. RWTH-PHOENIX-Weather and SIGNUM datasets were used to evaluate this approach and WER of 22.86% and 2.80% were reported, respectively. A modified version of LSTM was used for Indian SL recognition by Mittal et al. (2019). This approach was evaluated on 942 sentences collected using LMC and accuracies of 72.3% and 89.5% were reported on the sentence and word levels, respectively. A pseudo-supervised learning was used by Pei et al. (2019) for DGS recognition. The visual features were extracted using 3D-ResNet and CTC was used to map these features to the corresponding sentence text. To improve the feature representation and address the limitation of CTC objective function, 3D-ResNet was fine-tuned by feeding clip-level pseudo-labels and video clips. This technique was evaluated in RWTH-PHOENIX-Weather-2014 and a WER of 40.6% was obtained.

Wei et al. (2019) split the continuous sentence of CSL into segments and used 3D-ResNet for spatio-temporal sign video representation. Then, Bi-LSTM was used to model the temporal dependency across the sign videos. A global temporal pooling layer was used to summarize these segments into one feature vector to be fed into word-independent classifiers (WIC) and n-gram classifier (NGC) simultaneously. A dataset consisting of 100 sentences performed by 50 signers was used to evaluate this approach and a WER of 0.51 was reported. Zhou et al. (2019) employed 3D-CNN for feature extraction and bidirectional GRU (BGRU) for temporal modeling. CTC was used after this for sequence learning. RWTH-PHOENIX-Weather-2014 and CSL datasets were used to evaluate this approach and WER of 34.5% and 4.7% were reported, respectively. Pu et al. (2019) proposed an alignment module that was iteratively optimized for weakly supervised CSLR. This module consisted of 3D-ResNet for features extraction followed by BLSTM encoder and a decoder system consisting of CTC and LSTM. A soft-DTW was used between LSTM and CTC decoders for alignment. This approach was evaluated on RWTH-PHOENIX-Weather and CSL datasets yielding WER of 36.7% and 32.7%, respectively.

A reinforcement-based optimization was used by Zhang et al. to improve the recognition accuracy of the continuous sentences of DGS (Zhang et al., 2019a). This optimization technique was used for a model consisting of 3D-ResNet for feature extraction followed by neural machine translation model to translate signs into text sentences. RWTH-PHOENIX-Weather dataset was used to evaluate this model and a WER of 38.3% was reported. Yang et al. (2019) proposed Structured Feature Network (SF-Net) for CSLR recognition. The proposed model used different network designs to learn features at three levels: frame, gloss, and sentence. The features learnt at the frame level focus on manual and non-manual gestures in each frame. This level was followed by framing operation to capture the gloss level motions. The output of this level was fed into BLSTM and CTC network to capture the sentence context information. The reported WERs of applying this technique to CSL and RWTH-PHOENIX-Weather-2014 datasets were 3.8% and 34.9%, respectively.

6. Conclusions and future directions

Sign language recognition has come a long way from its initial starts in recognizing alphabets and digits. Current techniques of SLR are successful in dealing with dynamic signs that depend on hands and body motion. Moreover, current work rarely depends on using sensors or colored gloves that were used by earlier techniques. The availability of multi-modal devices that help in detecting and tracking signer body parts motivated researchers to abandon sensor-based technique in favor of vision-based approaches for SLR. To boost the recognition accuracy of sign language recognition systems, several researchers combined the manual gestures with non-manual gestures and promising results have been reported. This rapid increase in recognizing isolated signs was associated with building databases with different specifications to be used for sign language recognition.

More attention has been recently focused in developing recognition systems for continuous sentences of sign language. Though the tremendous research efforts, continuous sign language recognition and translation is still an open problem attracting more attention in recent years. It is far from being completely solved or becoming a commercial application as speech recognition systems (Luqman and Mahmoud, 2020). This can be attributed to the challenges associated with the sign language gestures and limitations of the currently available resources and techniques for segmentation and recognition. Continuous sign language sentences are naturally signed and movement epentheses between signs of two sentences are not similar and differ from one signer to another. In addition, the lack of an accurate temporal segmentation information for sentence signs makes finding an effective segmentation algorithm a critical issue. To address this issue, recent CSLR techniques perform automatic labeling and recognizing of sentences without the need for prior segmentation. HMM and deep learning technique are the most common techniques that have been used in the literature for CSLR. These techniques do not require pre-segmentation of the sentence and do automatic alignment between sentence segments and labels in addition to its high recognition capabilities.

The majority of the available datasets were recorded in a restricted environment, which may explain the high classification rates obtained by systems evaluated on these datasets. However, in order to reach to more reliable systems, the accuracy and applicability of SLR techniques should be evaluated on a more comprehensive large-scale dataset (Ibrahim et al., 2020). Almost all available SL datasets are limited in vocabulary size and number of signers which is more observable in CSLR compared with isolated signs datasets. Additionally, few SLR systems have been evaluated using signer-independent strategy. This type of evaluation is important to test the generalization of the recognition systems. This makes the construction of a larger annotated dataset covering the linguistics information of SL a highly demanded task especially for recent techniques requiring big data such as deep learning. The difficulty lies in the availability of expert signers and annotators and the high variation between signers trained to perform different signs.

Among other issues are the acquisition devices, multimodality fusion and applications of 3D technologies. While sensor-based devices, such as gloves, help in hands tracking and features extraction, requiring the signer to wear these devices during signing is mostly cumbersome and makes these devices less useful for natural spontaneous sign acquisition. Multimodal hardware devices, such as Kinect, have been efficiently deployed in signs collection. These devices help in body parts tracking and background removal. Sign language recognition is still in

its infancy stage though the effort made. Recently, deep learning has led to significant advances in many areas of computer vision including sign language recognition. As future perspectives, several challenges need to be investigated by researchers, including:

- Lack of datasets collected using mobile device camera although deaf nowadays depend mainly on mobile devices for communication and thoughts sharing. These databases can help in developing and evaluating real-time SLR systems.
- Lack of continuous sign language datasets for some sign languages such as ArSL and Turkish.
- To our knowledge, no research has been conducted on a collection of fingerspelling signs performed continuously. Most of the fingerspelling signs recognition systems are evaluated on hand's cropped images. However, recognizing the fingerspelling signs within video streams is more challenging.
- More research efforts are required for combining different modalities such as non-manual with manual gestures to further improve the accuracy of SLR systems due to the valuable information encoded with non-manual gestures.

CRediT authorship contribution statement

El-Sayed M. El-Alfy: Administration, Conceptualization, Methodology, Data curation, Visualization, Writing – editing, Critical review, Final revision. **Hamzah Luqman:** Data curation, Writing – original draft, Visualization, Writing – editing, Final revision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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